WIA 1002 DATA STRUCTURE

SEM 2, SESSION 2025/2026

Dr. Zainab Malik
zainabmalik@um.edu.my
K1 (OCC 1 – 6)

Dr. Uzair Ishtiaq
uzair@um.edu.my
K2 (OCC 7 – 12)

Ts. Dr. Siti Nurliana Binti Jamalai@Jamali
siti.nurliana@um.edu.my
K3 (OCC 13 – 15)

Home of the Bright, Land of the Brave
Di Sini Bermulanya Pintar, Tanah Tumpahnya Berani

UNIVERSITI MALAYA

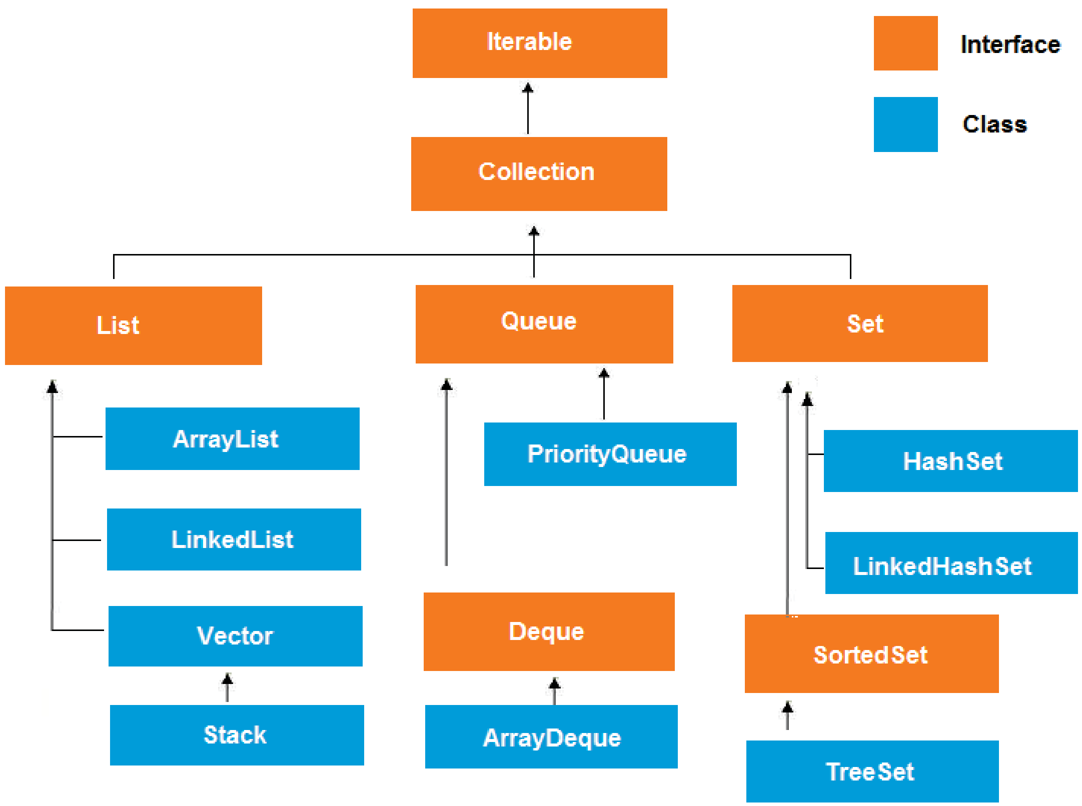
www.um.edu.my

Queue

- Recap
- Introduction
- Implementation
- Priority Queue

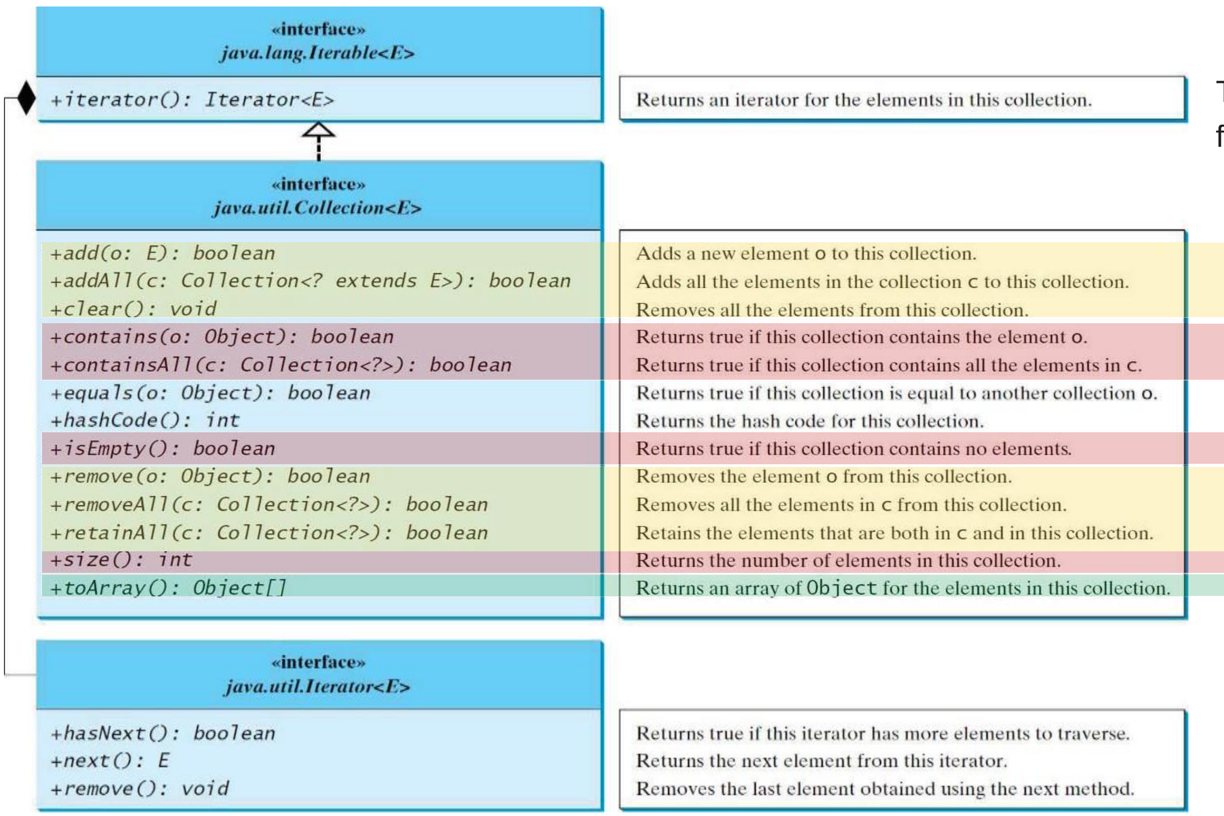
Recap

Java Collection Framework Hierarchy



The Collection Interface

The Collection interface is the root interface for manipulating a collection of objects.



Basic operations
Query Operations
Representation

Introduction

Queue

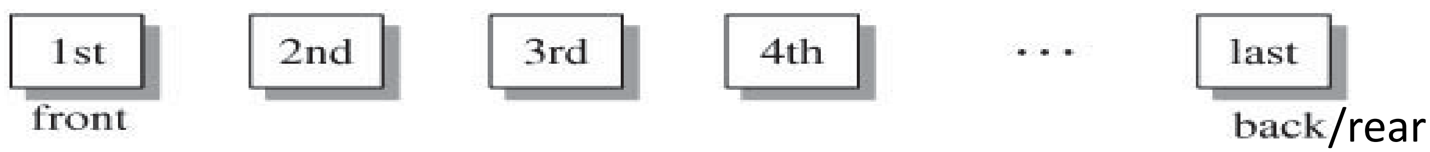
- A waiting line at a grocery store or a bank is a model of a queue.

A **B** **C** **D**

Which is first in this queue? Which is last?

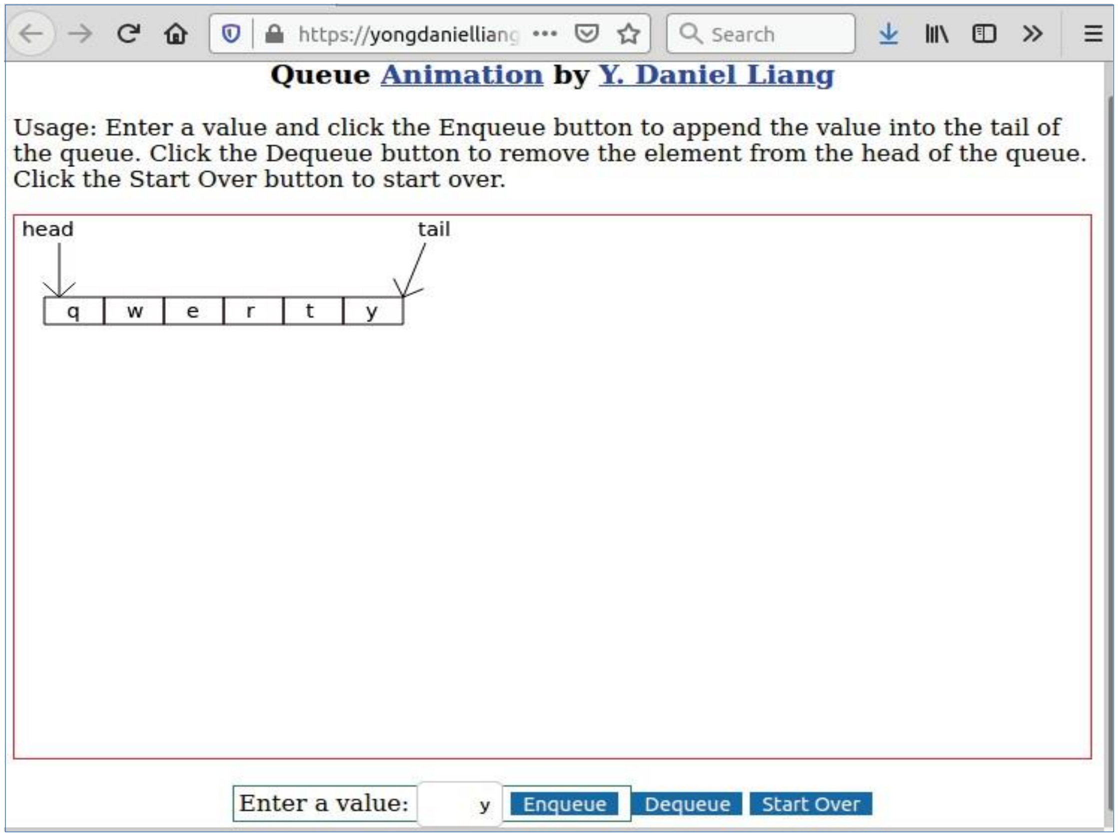
Queue

- A queue represents a waiting list.
- A queue can be viewed as a special type of **list**, where the elements are inserted into the end (back/tail) of the queue, and are **accessed** and **deleted** from the beginning (front/head) of the queue.



Queue

- An item **removed** from the queue is the **first** element that was added into the queue. A queue has **FIFO** (first-in-first-out) ordering.



https://yongdanielliang.github.io/animation/web/Queue.html

Check Point

1. Where is element inserted and deleted from a queue?
2. What is this ordering called?

Implementation

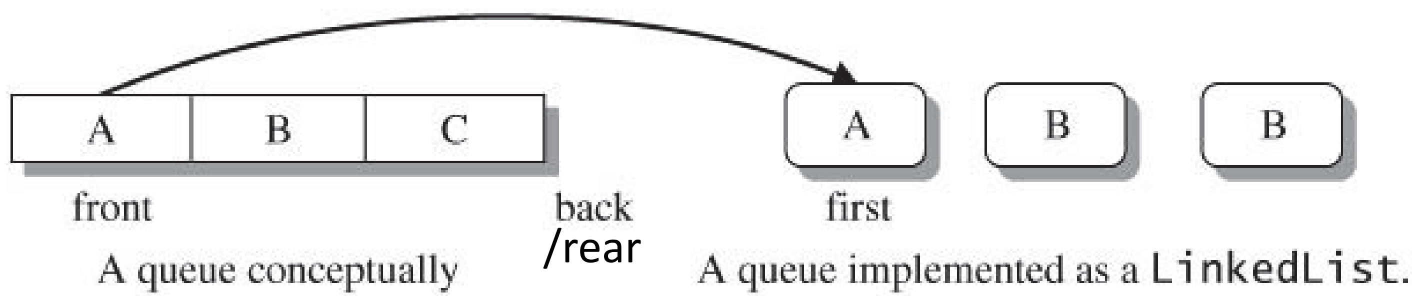
Methods/Operations in Queue

@<u>En</u>queue() → <u>en</u>(in) / add

@<u>De</u>queue()→ <u>d</u><u>e</u>lete

Implementing Queue

- Since deletions are made at the beginning of the list, it is more efficient to implement a queue using a linked list than an array list.
- Using LinkedList to implement Queue



Implementing Queue

2 ways to design the queue class using LinkedList:

- Using inheritance: You can define a queue (GenericQueue) class by extending LinkedList class
- Using composition: You can define an linked list as a data field in the queue(GenericQueue) class

Implementing Queue

(a) Using inheritance

(b) Using composition

GenericQueue<E> Class

- GenericQueue class using a LinkedList & composition approach.

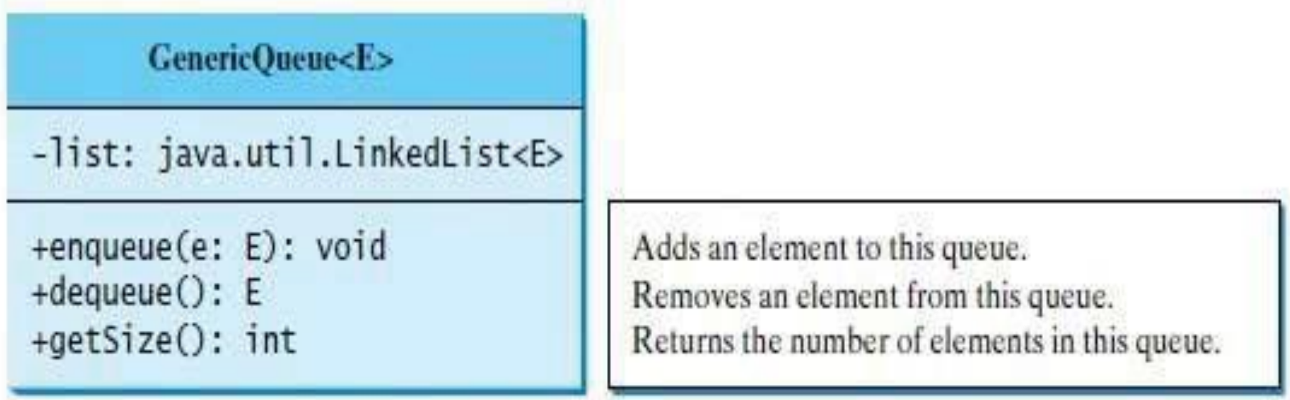


FIGURE 24.22 GenericQueue uses a linked list to provide a first-in, first-out data structure.

GenericQueue<E> Class

LISTING 24.7 GenericQueue.java

```java
public class GenericQueue<E> {
  private java.util.LinkedList<E> list
    = new java.util.LinkedList<>();

  public void enqueue(E e) {
    list.addLast(e);
  }

  public E dequeue() {
    return list.removeFirst();
  }

  public int getSize() {
    return list.size();
  }

  @Override
  public String toString() {
    return "Queue: " + list.toString();
  }
}
```

https://docs.oracle.com/javase/8/docs/api/java/util/LinkedList.html

Test GenericQueue class

```java
        // Create a queue
        GenericQueue<String> queue = new GenericQueue<>();

        // Add elements to the queue
        queue.enqueue("Tom"); // Add it to the queue
        System.out.println("(7) " + queue);

        queue.enqueue("Susan"); // Add it to the queue
        System.out.println("(8) " + queue);

        queue.enqueue("Kim"); // Add it to the queue
        queue.enqueue("Michael"); // Add it to the queue
        System.out.println("(9) " + queue);

        // Remove elements from the queue
        System.out.println("(10) " + queue.dequeue());
        System.out.println("(11) " + queue.dequeue());
        System.out.println("(12) " + queue);
    }
}
```

```
(7) Queue: [Tom]
(8) Queue: [Tom, Susan]
(9) Queue: [Tom, Susan, Kim, Michael]
(10) Tom
(11) Susan
(12) Queue: [Kim, Michael]
```

Check Point

1. Which type of data structure is best to use to implement Queue?
2. What are the 2 important operations for Queue?

Priority Queue

Priority Queue

- A **regular queue** is a **first-in and first-out (FIFO)** data structure. Elements are appended to the end of the queue and are removed from the beginning of the queue.

- In a **priority queue**, **elements are assigned with priorities.** When accessing elements, the element with the **highest priority is removed first**.

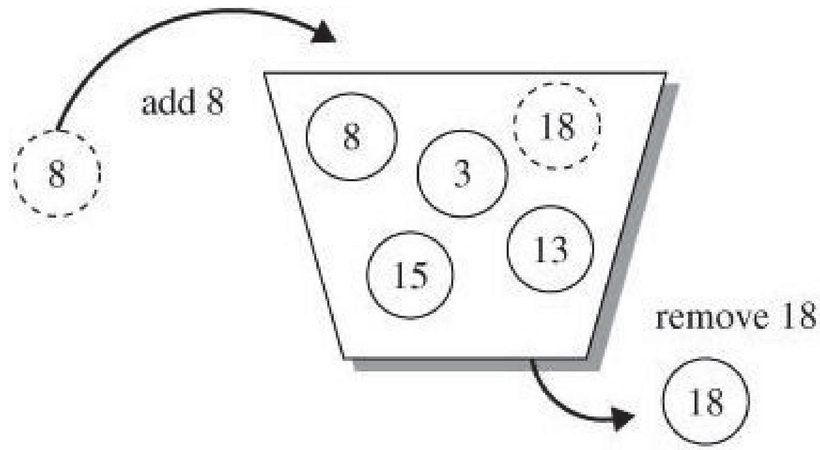
Priority Queue

- A priority queue has a **largest-in, first-out** behavior (however you need to double check whether the larger the higher the priority, or the smaller the higher the priority).
- For example, the emergency room in a hospital assigns priority numbers to patients; the patient with the highest priority is treated first. The assumption here the higher the number the higher the priority.

Priority Queue

- A priority queue is a collection in which all elements have a comparison (priority) ordering.
- It provides only simple access and update operations where a deletion always removes the element of highest priority.

Priority Queue



- In the above diagram, the assumption is the higher the number, the higher the priority.

The PriorityQueue Class from Java API

- import java.util.PriorityQueue Implemented interface java.util.Queue<E>
- https://docs.oracle.com/javase/8/docs/api/java/util/PriorityQueue.html

Constructors

Constructor and Description
`PriorityQueue()` Creates a `PriorityQueue` with the default initial capacity (11) that orders its elements according to their **natural ordering**.
`PriorityQueue(Collection<? extends E> c)` Creates a `PriorityQueue` containing the elements in the specified collection.
`PriorityQueue(int initialCapacity)` Creates a `PriorityQueue` with the specified initial capacity that orders its elements according to their **natural ordering**.
`PriorityQueue(int initialCapacity, Comparator<? super E> comparator)` Creates a `PriorityQueue` with the specified initial capacity that orders its elements according to the specified comparator.
`PriorityQueue(PriorityQueue<? extends E> c)` Creates a `PriorityQueue` containing the elements in the specified priority queue.
`PriorityQueue(SortedSet<? extends E> c)` Creates a `PriorityQueue` containing the elements in the specified sorted set.

PriorityQueue

- To add an element :
  - offer(E e) [returns boolean]
- To remove an object:
  - remove (Object o) [returns boolean]
- To retrieve, but not remove the head of the PriorityQueue:
  - peek () [returns E]
- To retrieve and remove the head of the PriorityQueue:
  - poll () [returns E]

PriorityQueue

- To clear the PriorityQueue:
  - clear() [void]
- To check the size of the Priority Queue :
  - size() [returns integer]
- To check whether object o is in the PriorityQueue:
  - contains (Object o) [returns boolean]

PriorityQueue Example 1

```java
import java.util.*;

public class PriorityQueueDemo {
  public static void main(String[] args) {
    PriorityQueue<String> queue1 = new PriorityQueue<>();
    queue1.offer("Oklahoma");
    queue1.offer("Indiana");
    queue1.offer("Georgia");
    queue1.offer("Texas");
    System.out.println("Priority queue using Comparable:");

    while (queue1.size() > 0) {
      System.out.print(queue1.poll() + " ");
    }

    PriorityQueue<String> queue2
      = new PriorityQueue<>(4, Collections.reverseOrder());
    queue2.offer("Oklahoma");
    queue2.offer("Indiana");
    queue2.offer("Georgia");
    queue2.offer("Texas");
    System.out.println("\nPriority queue using Comparator:");
    while (queue2.size() > 0) {
        System.out.print(queue2.poll() + " ");
    }
    System.out.println();
  }
}
```

```
Priority queue using Comparable:
Georgia Indiana Oklahoma Texas
Priority queue using Comparator:
Texas Oklahoma Indiana Georgia
```

PriorityQueue Example 2

```java
public class Customer implements Comparable<Customer> {
    private Integer id;
    private String name;

    public Customer(Integer id, String name) {
        this.id = id;
        this.name = name;
    }

    public Integer getID() {
      return id;
   }
   public void setID(Integer id) {
      this.id = id;
   }
    public String getName() {
      return name;
   }
   public void setName(String name) {
      this.name = name;
   }

    @Override
    public int compareTo(Customer c) {
        return this.getID().compareTo(c.getID());
    }

    @Override
    public String toString() {
        return "Customer [ id=" + id + ", name=" + name + " ]" ;
    }
}
```

Create a class Customer

PriorityQueue Example 2

```java
import java.util.*;

public class PriorityQueue2 {
  public static void main(String[] args) {
    PriorityQueue<Customer> customerQueue
      = new PriorityQueue<>(Collections.reverseOrder());

    customerQueue.add(new Customer(3, "Donald" ));
    customerQueue.add(new Customer(1, "Chong" ));
    customerQueue.add(new Customer(2, "Ali" ));
    customerQueue.add(new Customer(4, "Bala" ));

    Customer c = customerQueue.peek();
    if (c!=null) {
       System.out.println(c.getName() + " is in queue");
       while ((c = customerQueue.poll())!=null)
          System.out.println(c);
    }
     System.out.println();
  }
}
```

```
Bala is in the queue
Customer [ id=4, name=Bala ]
Customer [ id=3, name=Donald ]
Customer [ id=2, name=Ali ]
Customer [ id=1, name=Chong ]
```

Exercise

- Implement a generic Queue using array
- Implement a generic Queue using ArrayList
- Hints: you can name the class ArrayQueue<E> and provide the implementation for the following methods:
  - public ArrayQueue()
  - Public ArrayQueue(int initial)
  - public void enqueue(E e)
  - public E dequeue()
  - public E getElement()
  - public boolean isEmpty()
  - public int size()
  - public void resize()

Reference

- Chapter 19 and 24, Liang, Introduction to Java Programming, 10th Edition, Global Edition, Pearson, 2015